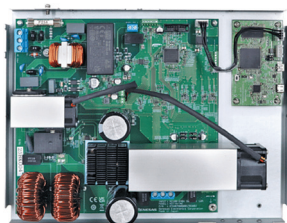


units (ODUs) for inverter air-conditioners. It combines motor control, power electronics, and system communication in one setup. The system controls both the compressor motor and the condenser fan



motor together, allowing testing of air-conditioner operation. It uses sensorless control methods to run PMSM motors at both low and high speeds, helping to improve energy efficiency, reduce noise, and increase reliability. The design also includes a power stage with an inverter and power factor correction (PFC) circuits to improve power efficiency and reduce electrical noise from the AC supply. Safety features such as overcurrent, overvoltage, and overheating protection are included. The platform also supports connections for temperature sensors and valves, as well as communication with the indoor unit, allowing you to test the complete air-conditioning system, including the motor control section.

Applications. The design is used in inverter air conditioners, heat pumps, HVAC outdoor units, compressor and fan motor control, variable-speed drives, and energy-efficient cooling systems.

Link. <https://www.electronicshobby.com/electronics-projects/reference-design-for-air-conditioner-outdoor-units>

OEM Brand. Renesas

SMPS for air-conditioners

This is a power supply design for home appliances and HVAC systems that need standby power. It operates with an AC input range of 85V AC-265V AC and provides two isolated DC outputs, 12V and 5V, for control boards, relays, sensors,

and communication circuits. The system uses an integrated power controller with a built-in high-voltage switch, thereby reducing external components, simplifying PCB design, improving reliability, and reducing development time. To



reduce standby power consumption, the design uses burst-mode control during light-load operation. The input stage includes an EMI filter, bridge rectifier, and bulk capacitor to reduce electrical noise and store energy, while frequency jittering spreads noise across frequencies to improve EMI performance and reduce filter size. Output voltage regulation is done through an optocoupler feedback circuit to maintain a stable voltage during load changes. The design also includes overcurrent, overload, undervoltage, overvoltage, and short-circuit protection with automatic restart for safe operation.

Applications. The reference design is suitable for HVAC systems

Link. <https://www.electronicshobby.com/electronics-projects/smcs-reference-design-for-air-conditioners>

OEM Brand. Infineon Technologies

HVAC control

This reference platform helps you build car climate control systems by integrating motor control, power management, sensors, and user controls into a single system. It uses a special automotive microcontroller that works efficiently and responds quickly in real time. The platform can control various motors used in car air-conditioning systems, such as those for moving air vents, actuators, and blower fans. Its design is divided into two parts: one board manages system control, sensors, and user inputs, while the

other board controls the motors and power functions, making the system easier to customise, test, and expand for different vehicles. The platform also uses sensors to detect temperature, humidity, light, and air



quality inside the car, helping the system automatically adjust airflow and cooling to improve comfort and reduce energy use. It can support displays and controls, such as buttons, rotary knobs, and touchscreens, so users can easily change settings such as temperature and fan speed. In addition, the software allows you to adjust and customise automatic climate control features based on the vehicle's cabin size and design.

Applications. The reference design is suitable for HVAC applications.

Link. <https://www.electronicshobby.com/electronics-projects/hvac-control-reference-design>

OEM Brand. NXP Semiconductor

These designs demonstrate how modern air-conditioning and HVAC systems are becoming more compact, intelligent, energy-efficient, and reliable through advanced electronic control and power-management technologies. By integrating motor control, inverter systems, sensorless operation, communication interfaces, safety protection, and efficient power conversion into unified platforms, these designs simplify development, improve system performance, and reduce energy consumption. **EFY**

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